

Problem Statement and Solution Requirements

Date	01 October 2026
Team ID	6ab22737f32d8c9d8b225a8a
Project Name	Implement Client Script & UI Policy (Incident)
Maximum Marks	4 Marks

Problem Statement

Users who request services through manual or disconnected channels are not told whether their request was received, who is handling it, or when it will be completed. Records are scattered, providers receive unclear instructions, and inventory is not updated in step with fulfilled requests. This project provides one platform to submit, validate, fulfil and track service requests.

Define Functional and Non-Functional Requirements

The requirements below define exactly what the solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (what the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (how the system should *be*).

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High / Medium / Low)
1	Authentication	Users register and log in with email or mobile number and password, verified by a one-time password (OTP). Sessions expire after inactivity.	High
2	Authorization levels	Three roles: User (submit and track own requests), Service Provider (view and update assigned requests only), Admin (manage users, providers, all requests and reports).	High
3	External interfaces	Integration with SMS/email gateway for notifications, a payment gateway for service charges, and the Service Provider system or app for sending service orders and receiving status responses.	High
4	Transactions processing	Each request is validated, stored in the Request Datastore, assigned to a provider, and updated through each status (Submitted, Assigned, In Progress, Completed). Payments are processed through the gateway and recorded against the request.	High
5	Reporting	Dashboards and exportable reports (PDF/CSV) for requests by status, provider performance, average completion time, inventory usage, and request history per user.	Medium
6	Business rules, etc.	Duplicate requests are rejected; mandatory fields must be complete; requests are auto-assigned by service type and provider availability; users are notified at every status change; completed requests cannot be edited.	High

S.No	Requirement Category	Requirement Description	Priority (High / Medium / Low)
7	Compliance to laws or regulations	Personal data is handled under India's Digital Personal Data Protection (DPDP) Act, 2023: user consent is captured and users can request data correction or deletion.	Medium
8	Other	Inventory is updated automatically when a request uses parts or materials, and users can give feedback and a rating after completion.	Low

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	Pages and status checks must respond quickly, and acknowledgements must be immediate.	Page loads in < 2 seconds; acknowledgement sent in < 5 seconds
2	Scalability	The system must handle growth in users, requests and stored data without redesign.	Supports up to 5,000 concurrent users and 100,000 requests per month
3	Security & Data Privacy	Data is encrypted in transit and at rest, access is role-based, and vulnerabilities are tested regularly.	HTTPS (TLS 1.2+) and AES-256 encryption at rest; quarterly security testing
4	Reliability & Availability	The system must remain available with automatic backups and recovery from failures.	99.9% uptime per month; daily backups; recovery within 4 hours
5	Usability & Accessibility	Simple, mobile-friendly interface that a first-time user can use without training.	Complies with WCAG 2.1 AA; a request can be submitted in < 3 minutes
6	Other	Maintainability and portability: modular code, documented APIs, and support for major browsers and Android/iOS devices.	Updates deployed without downtime; works on latest 2 versions of major browsers